CHAPTER III

Groups, Bundles, and Groupoids

1 Lie Groups and Lie Algebras

1.1 Lie Algebra Actions and the Momentum Map

This section describes the main class of examples of Poisson manifolds that are not symplectic. Here G is a Lie group, $\mathfrak{g}$ its Lie algebra, and $\mathfrak{g}^*$ is the dual of $\mathfrak{g}$.

Definition 1.1.1. *The* $(\pm)$ **Lie–Poisson structure** *on* $\mathfrak{g}^*$ *is given by the Poisson bracket*

$$\{f, g\}_{\pm}(\theta) := \pm\theta([df_\theta, dg_\theta]); \tag{1.1}$$

here the differential df_θ *of* $f \in C^\infty(\mathfrak{g}^*, \mathbb{R})$ *at* $\theta \in \mathfrak{g}^*$*, which is a linear map from* $T_\theta\mathfrak{g}^* \simeq \mathfrak{g}^*$ *to* $\mathbb{R}$*, is identified with an element of* $\mathfrak{g} \simeq \mathfrak{g}^{**}$*, so that the right-hand side of (1.1) is the Lie bracket in* $\mathfrak{g}$*. The space* $\mathfrak{g}^*$ *equipped with the Poisson bracket (1.1) is denoted by* $\mathfrak{g}^*_\pm$*; hence* $C^\infty(\mathfrak{g}^*_\pm, \mathbb{R})$ *stands for the associated Poisson algebra.*

For $X \in \mathfrak{g}$ let $\tilde{X}$ be the linear function on $\mathfrak{g}^*$ defined by $\tilde{X}(\theta) := \theta(X)$; clearly $\tilde{X} \in C^\infty(\mathfrak{g}^*, \mathbb{R})$. From (1.1) one then obtains

$$\{\tilde{X}, \tilde{Y}\}_{\pm} = \pm\widetilde{[X, Y]}; \tag{1.2}$$

cf. II.(2.10) and surrounding text. In fact, the Poisson structure is determined by the special case (1.2). For let $\{T_a\}$ be a basis of $\mathfrak{g}$, with $[T_a, T_b] = C^c_{ab}T_c$, and dual basis $\{\omega^a\}$ of $\mathfrak{g}^*$, defined by $\omega^a(T_b) = \delta^a_b$. We then have global coordinates θ_a on $\mathfrak{g}^*$ (so that $\theta = \theta_a\omega^a$), and $\tilde{T}_a$ is simply the coordinate function θ_a. We know that $\{f, g\}$ depends linearly on df and dg; cf. I.(2.4). Since $df = (\partial f/\partial\theta_a)d\tilde{T}_a$, the claim follows. Evidently, $\{\tilde{T}_a, \tilde{T}_b\}_{\pm} = \pm C^c_{ab}\tilde{T}_c$, or $\{\theta_a, \theta_b\}_{\pm} = \pm C^c_{ab}\theta_c$. Thus, omitting the argument θ, (1.1) may be written as

$$\{f, g\}_{\pm} = \pm C^c_{ab}\theta_c \frac{\partial f}{\partial \theta_a}\frac{\partial g}{\partial \theta_b}. \tag{1.3}$$

We now look at the representation theory of $C^\infty(\mathfrak{g}^*_-, \mathbb{R})$ (in the sense of I.2.6.1). By Corollary I.2.6.5 a representation of $C^\infty(\mathfrak{g}^*_-, \mathbb{R})$ corresponds to a symplectic manifold S and a smooth map $J : S \to \mathfrak{g}^*$, such that $\{J^*f, J^*g\}_S = J^*\{f, g\}_-$ for all $f, g \in C^\infty(\mathfrak{g}^*, \mathbb{R})$. That is, $J : S \to \mathfrak{g}^*_-$ is a Poisson map. Let

$$J_X := J^*\tilde{X}, \tag{1.4}$$

which is in $C^\infty(S, \mathbb{R})$; in other words, $J_X(\sigma) := (J(\sigma))(X)$.

Proposition 1.1.2. *A smooth map $J : S \to \mathfrak{g}^*_-$ is Poisson iff $\{J_X, J_Y\}_S = -J_{[X,Y]}$ for all $X, Y \in \mathfrak{g}$.*

If B^S is the Poisson tensor on S, then $\{J^*f, J^*g\}_S = B^S_\sigma(J^*df, J^*dg\}$. As in the previous paragraph, this implies that

$$\{J^*f, J^*g\}_S(\sigma) = \frac{\partial f}{\partial \theta_a}(J(\sigma))\frac{\partial g}{\partial \theta_b}(J(\sigma))\{J^*\tilde{T}_a, J^*\tilde{T}_b\}(\sigma). \tag{1.5}$$

By assumption $\{J^*\tilde{T}_a, J^*\tilde{T}_b\}_S(\sigma) = -C^c_{ab}\tilde{T}_c(J(\sigma))$, so that the right-hand side of (1.5) is $\{f, g\}_-(J(\sigma))$. ∎

Define $\xi_X := \xi_{J_X}$, which is the Hamiltonian vector field of J_X. Assuming that J is indeed a Poisson morphism, the Jacobi identity on the Poisson bracket of S (or (I.2.9)) implies that

$$[\xi_X, \xi_Y] = -\xi_{[X,Y]}; \tag{1.6}$$

here the left-hand side contains the commutator of vector fields, whereas on the right-hand side [,] stands for the Lie bracket in $\mathfrak{g}$.

Let us refer to a linear map $X \mapsto \xi_X$ of $\mathfrak{g}$ into the space of vector fields $\Gamma(TS)$ on S satisfying (1.6) as a **$\mathfrak{g}$-action** on S. Here $\Gamma(TS)$ may be regarded as the Lie algebra of the diffeomorphism group of S, whose Lie bracket is minus the commutator (cf. I.3.3), so (1.6) corresponds to a Lie algebra homomorphism as appropriate. When various $\mathfrak{g}$-actions play a role we sometimes write ξ^S_X for ξ_X. If $\{T_a\}$ is a basis of $\mathfrak{g}$, we abbreviate $\xi_a := \xi_{T_a}$. One speaks of a **Poisson $\mathfrak{g}$-action** when S is a Poisson manifold and $L_{\xi_X}B = 0$ for all $X \in \mathfrak{g}$, where B is the Poisson tensor. When (S, ω) is symplectic, which is the only case we shall consider in the context of $\mathfrak{g}$-actions, this condition is equivalent to $L_{\xi_X}\omega = 0$ for all X.

We infer from I.(2.10) that a representation of $C^\infty(\mathfrak{g}^*_-, \mathbb{R})$ on S leads to a Poisson $\mathfrak{g}$-action on S. Conversely, one may ask whether a given $\mathfrak{g}$-action on a symplectic manifold S is related to a representation of $C^\infty(\mathfrak{g}^*_-, \mathbb{R})$ on S.

Definition 1.1.3. *A* **momentum map** *for a $\mathfrak{g}$-action $X \mapsto \xi_X$ on S is a map $J : S \to \mathfrak{g}^*$ for which*

$$\xi_{J_X} = \xi_X \tag{1.7}$$

for all $X \in \mathfrak{g}$; here J_X is defined by (1.4).

This definition applies to general Poisson manifolds, but we will use it only when S is symplectic. We will occasionally write J_a for J_{T_a}. A **Hamiltonian $\mathfrak{g}$-action** on a symplectic manifold is a $\mathfrak{g}$-action given by a momentum map as in (1.7). It is clear from (1.7) and I.(2.10) that a Hamiltonian $\mathfrak{g}$-action is Poisson. When a momentum map J exists, (1.7) is equivalent to

$$i_{\xi_X}\omega = dJ_X \tag{1.8}$$

for all $X \in \mathfrak{g}$; to prove this, contract both sides with ξ_f (which is the most general type of local vector field, since S is symplectic) and use I.(2.19).

Conversely, when a $\mathfrak{g}$-action is Poisson, the properties $d\omega = 0$ and $L_{\xi_X}\omega = 0$ and the identity $L_\xi = i_\xi d + di_\xi$ imply $di_{\xi_X}\omega = 0$, so that by Poincaré's lemma a function J_X satisfying (1.8) must exist at least locally.

Proposition 1.1.4. *Sufficient conditions for the existence of a momentum map for a Poisson $\mathfrak{g}$-action on a symplectic manifold (S, ω) are $H^1_{\mathrm{dR}}(S, \mathbb{R}) = 0$ or $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ (equivalently, $H^1(\mathfrak{g}, \mathbb{R}) = 0$).*

Here $H^1_{\mathrm{dR}}(S, \mathbb{R}) := Z^1_{\mathrm{dR}}(S, \mathbb{R})/B^1_{\mathrm{dR}}(S, \mathbb{R})$ is the first de Rham cohomology group of S; recall that $Z^1_{\mathrm{dR}}(S, \mathbb{R})$ and $B^1_{\mathrm{dR}}(S, \mathbb{R})$ are the spaces of all closed and all exact 1-forms on S, respectively. The sufficiency of the condition $H^1_{\mathrm{dR}}(S, \mathbb{R}) = 0$ is evident from the paragraph preceding this proposition.

The vector space $H^1(\mathfrak{g}, \mathbb{R})$ is the **first cohomology group of** $\mathfrak{g}$; since $B^1(\mathfrak{g}, \mathbb{R})$ is identically zero, $H^1(\mathfrak{g}, \mathbb{R})$ is defined as the subspace $Z^1(\mathfrak{g}, \mathbb{R}) \subset \mathfrak{g}^*$ of all θ for which $\theta([X, Y]) = 0$ for all $X, Y \in \mathfrak{g}$. The equivalence between the conditions $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ and $H^1(\mathfrak{g}, \mathbb{R}) = 0$ is obvious.

If $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, then an arbitrary $X \in \mathfrak{g}$ can be written as $X = \sum_i X_i$ with $X_i = [Y_i, Z_i]$ for appropriate $Y_i, Z_i \in \mathfrak{g}$. If $X = [Y, Z]$, we choose $J_X = \omega(\xi_Y, \xi_Z)$, which, by an elementary calculation, using (1.6) and $d\omega = 0$, satisfies (1.8) and hence (1.7). For arbitrary X we define J_X by linear extension of this expression. ∎

The existence of a momentum map J in itself does not imply that J preserves the Poisson bracket. To detect the extent to which it does we define a function Γ on $\mathfrak{g} \times \mathfrak{g} \times S$ by

$$\{J_X, J_Y\}_S = -J_{[X,Y]} - \Gamma(X, Y). \tag{1.9}$$

It is clear that Γ is bilinear and antisymmetric in X, Y. Taking the Poisson bracket of both sides of (1.9) with an arbitrary $f \in C^\infty(S)$, and using (1.7), (1.6), and the Jacobi identity, we infer that $\{\Gamma(X, Y), f\} = 0$ for all X, Y. Since S is symplectic, this shows that Γ does not depend on its argument in S. A bilinear function $\Gamma :$ $\mathfrak{g} \otimes \mathfrak{g} \to \mathbb{R}$ satisfying

$$\Gamma(X, Y) = -\Gamma(Y, X), \tag{1.10}$$

$$\Gamma(X, [Y, Z]) + \Gamma(Z, [X, Y]) + \Gamma(Y, [Z, X]) = 0 \tag{1.11}$$

is called a **2-cocycle** on $\mathfrak{g}$ (with values in $\mathbb{R}$). The space of all 2-cocycles on $\mathfrak{g}$ is denoted by $Z^2(\mathfrak{g}, \mathbb{R})$. It follows from the Jacobi identity on both $\{\ ,\ \}_S$ and $[\ ,\ ]$ that Γ as defined in (1.9) is indeed an element of $Z^2(\mathfrak{g}, \mathbb{R})$.

We are now motivated to define a modified Lie–Poisson bracket on $\mathfrak{g}^*$ by

$$\{f, g\}^{\Gamma}_{\pm} := \{f, g\}_{\pm} \pm \Gamma(df, dg); \tag{1.12}$$

this is indeed a Poisson bracket on account of (1.11). Generalizing (1.3), in coordinates one has

$$\{f, g\}^{\Gamma}_{\pm} = \pm\left(C^c_{ab}\theta_c + \Gamma_{ab}\right)\frac{\partial f}{\partial\theta_a}\frac{\partial g}{\partial\theta_b}, \tag{1.13}$$

$$\Gamma_{ab} := \Gamma(T_a, T_b). \tag{1.14}$$

As for $\Gamma = 0$, one shows that this modified Poisson bracket is determined by the special case

$$\{\tilde{X}, \tilde{Y}\}^{\Gamma}_{\pm} = \pm\left(\widetilde{[X, Y]} + \Gamma(X, Y)\right). \tag{1.15}$$

Definition 1.1.5. *The space $\mathfrak{g}^*$ equipped with the Poisson bracket (1.13) is denoted by $\mathfrak{g}^*_{(\Gamma)\pm}$; we sometimes write $C^\infty_\Gamma(\mathfrak{g}^*_\pm)$ for the associated Poisson algebra.*

Generalizing Proposition 1.1.2, one easily proves

Proposition 1.1.6. *A smooth map $J : S \to \mathfrak{g}^*_{(\Gamma)-}$ is Poisson iff (1.9) holds.*

The essence of the preceding discussion may now be summarized as follows.

Theorem 1.1.7. *There is a bijective correspondence between representations π of $C^\infty_\Gamma(\mathfrak{g}^*_-)$ (in the sense of I.2.6.1) and Hamiltonian $\mathfrak{g}$-actions with given complete momentum map and associated 2-cocycle Γ. Given $\pi : C^\infty_\Gamma(\mathfrak{g}^*_-) \to C^\infty(S)$ one constructs a Poisson map $J : S \to \mathfrak{g}^*_{(\Gamma)-}$ by I.2.6.5, and subsequently defines the $\mathfrak{g}$-action $X \mapsto \xi_X$ by (1.7). Conversely, given a $\mathfrak{g}$-action with associated complete momentum map J (yielding a 2-cocycle Γ), one puts $\pi = J^*$.*

The fact that $X \mapsto \xi_X$ is indeed a $\mathfrak{g}$-action follows from the argument leading to (1.6); even when $\Gamma \neq 0$ the additional term in (1.12) is a constant function, so that the Jacobi identity on the Poisson bracket still implies (1.6). It is Poisson by I.(2.10), and Hamiltonian with 2-cocycle Γ by construction. In the converse the fact that J^* is a representation is immediate from 1.1.6. ■

A **strongly Hamiltonian $\mathfrak{g}$-action** is a Hamiltonian $\mathfrak{g}$-action possessing a momentum map $J : S \to \mathfrak{g}^*_-$ that is Poisson; in other words, there exists a J for which $\Gamma = 0$ in (1.9). A Hamiltonian $\mathfrak{g}$-action with 2-cocycle Γ may alternatively be described as a strongly Hamiltonian action of a certain Γ-dependent Lie algebra containing $\mathfrak{g}$.

Definition 1.1.8. *The* **central extension** $\mathfrak{g}_\Gamma$ *of a Lie algebra $\mathfrak{g}$ by $\mathbb{R}$ relative to some $\Gamma \in Z^2(\mathfrak{g}, \mathbb{R})$ is $\mathfrak{g}_\Gamma := \mathfrak{g} \oplus \mathbb{R}$ as a vector space, equipped with the Lie bracket*

$$[X, Y]_\Gamma = [X, Y] + \Gamma(X, Y)T_0; \tag{1.16}$$

$$[X, T_0]_\Gamma = 0 \tag{1.17}$$

for $X, Y \in \mathfrak{g}$, and T_0 a basis vector of the extension $\mathbb{R}$.

The Jacobi identity for $[\,,\,]_\Gamma$ is a consequence of (1.11). We have an embedding $\iota : \mathfrak{g} \hookrightarrow \mathfrak{g}_\Gamma$ by $\iota(X) = X \dot{+} 0$ (which is not a Lie algebra homomorphism unless $\Gamma = 0$), as well as a quotient $\mathfrak{g} \simeq \mathfrak{g}_\Gamma/\mathbb{R}$ as Lie algebras.

Proposition 1.1.9. *There is a bijective correspondence between Poisson maps $J : S \to \mathfrak{g}^*_{(\Gamma)-}$ (or, equivalently, Hamiltonian $\mathfrak{g}$-actions with 2-cocycle Γ) and Poisson maps $J_\Gamma : S \to \mathfrak{g}_{\Gamma-}$ (or strongly Hamiltonian $\mathfrak{g}_\Gamma$-actions) in which $(J_\Gamma(\sigma))(T_0) = 1$ for all $\sigma \in S$ (equivalently, $\pi(\tilde{T}_0) = 1_S$, so that $\xi_{T_0} = 0$). This correspondence preserves irreducibility.*

Let ω^0 be the basis element in $\mathfrak{g}_\Gamma$ dual to T_0. Then $J_1 : \mathfrak{g}^*_{(\Gamma)-} \to \mathfrak{g}_{\Gamma-}$ given by $J_1(\theta) := \theta \dot{+} \omega^0$ is a Poisson map (where $\mathfrak{g}^*$ is embedded in $\mathfrak{g}^*_\Gamma$ as the annihilator of the extension $\mathbb{R}$); this follows from (1.15), (1.16), (1.2), and $J_1^* \tilde{T}_0 = 1_{\mathfrak{g}^*}$.

Given $J : S \to \mathfrak{g}^*_{(\Gamma)-}$, one constructs $J_\Gamma : S \to \mathfrak{g}_{\Gamma-}$ by $J_\Gamma := J_1 \circ J$. Conversely, when a given J_Γ is as stated, the equality $J^*_\Gamma(\tilde{T}_0) = 1_S$ and Proposition 1.1.6 imply that $J_X(\sigma) := (J_\Gamma)_{\iota(X)}(\sigma)$ with (1.4) is a Poisson map $J : S \to \mathfrak{g}^*_{(\Gamma)-}$. Finally, Definition I.2.6.6 and the fact that $\xi_{T_0} = 0$ lead to the last part of the proposition. ■

As a by-product of the proof we have

Proposition 1.1.10. *The canonical identification of $C^\infty(\mathfrak{g}^*_{\Gamma-})/\ker(J_1^*)$ with $C^\infty_\Gamma(\mathfrak{g}^*_-)$ is a Poisson isomorphism.*

This is immediate from the definitions and (1.13). ■

The theory so far has been concerned with a Hamiltonian $\mathfrak{g}$-action with given momentum map J. However, when some J exists, then any map $J' = J + \theta_0$, where $\theta_0 \in \mathfrak{g}^*$, is obviously a momentum map for the given $\mathfrak{g}$-action as well. Having found a particular J for which $\Gamma \neq 0$, when can we redefine $J \mapsto J'$ so as to make J' a Poisson morphism?

A 2-cocycle $\Gamma \in Z^2(\mathfrak{g}, \mathbb{R})$ is said to be **trivial** when

$$\Gamma(X, Y) = \theta_0([X, Y]) \tag{1.18}$$

for some $\theta_0 \in \mathfrak{g}^*$. The subspace of trivial 2-cocycles is called $B^2(\mathfrak{g}, \mathbb{R})$. The quotient $H^2(\mathfrak{g}, \mathbb{R}) := Z^2(\mathfrak{g}, \mathbb{R})/B^2(\mathfrak{g}, \mathbb{R})$ is the **second cohomology group of** $\mathfrak{g}$.

Proposition 1.1.11. *If a momentum map of a Hamiltonian $\mathfrak{g}$-action defines a 2-cocycle satisfying (1.18), then the $\mathfrak{g}$-action is strongly Hamiltonian. In particular, when $H^2(\mathfrak{g}, \mathbb{R}) = 0$ any Hamiltonian $\mathfrak{g}$-action is strongly Hamiltonian.*

Given (1.18), the redefined momentum map $J' = J + \theta_0 : S \to \mathfrak{g}^*_-$ is a Poisson map by (1.9). The condition $H^2(\mathfrak{g}, \mathbb{R}) = 0$ implies that any 2-cocycle Γ is given by (1.18). ■

Combining 1.1.11 and 1.1.4 we obtain

Corollary 1.1.12. *Let $H^1(\mathfrak{g},\mathbb{R}) = H^2(\mathfrak{g},\mathbb{R}) = 0$. Any Poisson $\mathfrak{g}$-action is strongly Hamiltonian, and is associated with a unique Poisson momentum map $J : S \to \mathfrak{g}^*_-$.*

If one has a Poisson momentum map J, one may still shift $J \mapsto J + \theta_0$, preserving the Poisson property, iff θ_0 annihilates $[\mathfrak{g},\mathfrak{g}]$; when $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$ this forces $\theta_0 = 0$. ■

The conditions $H^i(\mathfrak{g},\mathbb{R}) = 0, i = 1, 2$, are satisfied, for example, when G is semisimple. The consistency between 1.1.9 and 1.1.11 is guaranteed by

Proposition 1.1.13. *If $\Gamma(X, Y) = \theta_0([X, Y])$ for some $\theta_0 \in \mathfrak{g}^*$, then $\mathfrak{g}_\Gamma$ is isomorphic to the trivial extension $\mathfrak{g}_0 = \mathfrak{g} \oplus \mathbb{R}$ as a direct sum of Lie algebras. In particular, when $H^2(\mathfrak{g},\mathbb{R}) = 0$, any central extension of $\mathfrak{g}$ by $\mathbb{R}$ is trivial.*

If $\Gamma(X, Y) = \theta_0([X, Y])$, then $X \mapsto X + \theta_0(X)T_0$ for $X \in \mathfrak{g}$ and $T_0 \mapsto T_0$ is the desired isomorphism between $\mathfrak{g}_\Gamma$ and $\mathfrak{g}_0$. ■

1.2 Hamiltonian Group Actions

We will now relate $\mathfrak{g}$-actions to G-actions, where G is a Lie group with Lie algebra $\mathfrak{g}$. Recall that a (left) **action** L of a group G on a manifold S is a map $L : G \times S \to S$, satisfying $L(e,\sigma) = \sigma$ and $L(x, L(y,\sigma)) = L(xy,\sigma)$ for all $\sigma \in S$ and $x, y \in G$. If G is a Lie group, we assume that L is smooth, unless the contrary is explicitly stated. We write $L_x(\sigma) = x\sigma := L(x,\sigma)$.

Given a Lie group action, one defines a linear map $X \mapsto \xi_X$ by

$$\xi_X f(\sigma) := \frac{d}{dt} f(\mathrm{Exp}(tX)\sigma)_{|t=0}, \tag{1.19}$$

where $\mathrm{Exp} : \mathfrak{g} \to G$ is the usual exponential map. One sees that (1.6) holds, so that $X \mapsto \xi_X$ is a Lie algebra homomorphism from $\mathfrak{g}$ into the Lie algebra $\Gamma(TS)$ of $\mathrm{Diff}(S)$. We say that the ξ_X generate the G-action, and call ξ_X a **generator** defined by X.

Conversely, one may ask whether a representation of $C^\infty(\mathfrak{g}^*_-,\mathbb{R})$ on S is derived from a G-action, in which case the representation is called **integrable**. This question is partly answered by the following statement.

Theorem 1.2.1. *Let $X \mapsto \xi_X$ be a homomorphism as above, and suppose the flow of each ξ_X, $X \in \mathfrak{g}$, is complete (this is the case iff there is a basis $\{T_a\}$ of $\mathfrak{g}$ such that the flow of each ξ_a is complete). Then the ξ_X generate an action of the simply connected Lie group $\tilde{G}$ whose Lie algebra is $\mathfrak{g}$.*

The construction of the $\tilde{G}$-action is, of course, done with the flow of the generators; that is, if $\sigma \mapsto \sigma(t)$ is the flow generated by ξ_X, one puts $\mathrm{Exp}(tX) : \sigma \mapsto \sigma(t)$. Such one-parameter groups generate $\tilde{G}$ (which is connected), but it remains to check that one indeed obtains a smooth group action. □

Note that the statement about the basis is nontrivial, since in principle the sum of two complete vector fields may be incomplete. Clearly, the hypothesis is automatically satisfied when S is compact, or more generally when all ξ_X have compact support. In any case, if the theorem leads to a $\tilde{G}$-action and if no ξ_X is identically zero, there is a discrete normal subgroup $D_S \subset \tilde{G}$ such that D_S is the maximal subgroup of $\tilde{G}$ that acts trivially on S. If $G = \tilde{G}/D$ for some discrete central subgroup $D \subset \tilde{G}$ (recall that any Lie group with Lie algebra $\mathfrak{g}$ is of this form), then the ξ_X generate a G-action if $D \subseteq D_S$.

When the $\mathfrak{g}$-action associated to a G-action on a symplectic manifold S is Hamiltonian, one speaks of a **Hamiltonian group action.** Similarly, a **strongly Hamiltonian group action** is an action for which a momentum map $J : S \to \mathfrak{g}^*$ exists that is a Poisson map; cf. (1.19) and 1.1.3. It is immediate from the comment preceding I.2.3.5 that a Hamiltonian G-action automatically consists of Poisson maps. Further to this, the conditions for a G-action on a symplectic manifold to be (strongly) Hamiltonian are entirely determined by the properties of the associated $\mathfrak{g}$-action, and are therefore given by Propositions 1.1.4 and 1.1.11 and Corollary 1.1.12.

The Hamiltonian version of **Noether's theorem** is as follows.

Proposition 1.2.2. *Given a Hamiltonian G-action on a symplectic manifold S, when $h \in C^\infty(S, \mathbb{R})$ is G-invariant (i.e., $h(x\sigma) = h(\sigma)$ for all $x \in G$), each J_X is constant along the Hamiltonian flow lines of h.*

Putting $x = \text{Exp}(tX)$ in $h(x\sigma) = h(\sigma)$, evaluating d/dt at $t = 0$, and using (1.7) leads to $\{J_X, h\} = 0$, which by I.(2.8) and I.(2.11) implies the claim. ∎

In view of this, in physics the components J_X of the momentum map usually play the role of conserved charges.

A Hamiltonian G-action enjoys a certain equivariance property. Recall the **adjoint action** Ad of G on $\mathfrak{g}$, defined by $\text{Ad}(x)Y := xYx^{-1}$ (more precisely, if $Y = d\gamma(t)/dt|t = 0$, then $\text{Ad}(x)Y = dx\gamma(t)x^{-1}|t = 0$). The derived representation of $\mathfrak{g}$ is then given by $\text{ad}(X)Y = [X, Y]$ (where we simply write ad for the awkward dAd). The **coadjoint action** Co of G on $\mathfrak{g}^*$ is defined by $(\text{Co}(x)\theta)(Y) := \theta(\text{Ad}(x^{-1})Y)$, with derived action of $\mathfrak{g}$ on $\mathfrak{g}^*$ written as $\text{co} := d\text{Co}$. One has $\text{ad}(T_a)T_b = C^c_{ab}T_c$, whence

$$\text{co}(T_a)\theta_b = -C^c_{ab}\theta_c. \tag{1.20}$$

As a first application of these definitions we note:

Proposition 1.2.3. *The Lie–Poisson structure is invariant under the coadjoint action (in other words, the map* $\text{Co}(x)$ *is a Poisson map for each $x \in G$).*

By the comment following (1.2) it suffices to show that $\{\tilde{X} \circ \text{Co}(x), \tilde{Y} \circ \text{Co}(x)\}_\pm = \widetilde{\{\tilde{X}, \tilde{Y}\}_\pm} \circ \text{Co}(x)$ for all $X, Y \in \mathfrak{g}$ and $x \in G$. Since $\tilde{X} \circ \text{Co}(x) = \widetilde{\text{Ad}(x^{-1})X}$ etc., this is evident from the fact that the adjoint action is an automorphism of the Lie algebra. ∎

Given a choice J of a momentum map associated to a Hamiltonian G-action on S via the derived $\mathfrak{g}$-action, we define $\gamma : G \times S \to \mathfrak{g}^*$ by

$$\gamma(x, \sigma) = J(x\sigma) - \mathrm{Co}(x)J(\sigma). \tag{1.21}$$

Lemma 1.2.4. *The function γ is independent of σ, and satisfies*

$$\gamma(xy) = \gamma(x) + \mathrm{Co}(x)\gamma(y). \tag{1.22}$$

A smooth map $\gamma : G \to \mathfrak{g}^*$ with property (1.22) is called a **1-cocycle on G with values in $\mathfrak{g}^*$**; the space of such 1-cocycles is denoted by $Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$. The proof that (1.21) is independent of σ is similar to the argument after (1.9). For arbitrary $f \in C^\infty(S)$ we compute $\{J_Y \circ L_x, f\}_S$ for fixed $x \in G$ and $Y \in \mathfrak{g}$, and use (1.7) and the invariance of the Poisson bracket under L_x. This shows that $\{J_Y \circ L_x, f\}_S = \{J_{\mathrm{Ad}(x^{-1})Y}, f\}_S$. Since S is symplectic, γ_Y (in obvious notation) is therefore constant in σ for all Y. The property (1.22) is then immediate from the definition and the σ-independence of γ and the equality $\mathrm{Co}(xy) = \mathrm{Co}(x)\mathrm{Co}(y)$. ■

Corollary 1.2.5. *A momentum map J for a Hamiltonian G-action is equivariant with respect to the modified coadjoint G-action on $\mathfrak{g}^*$ defined by*

$$\mathrm{Co}^\gamma(x)\theta := \mathrm{Co}(x)\theta + \gamma(x), \tag{1.23}$$

where γ is given by (1.21). That is, $J \circ L_x = \mathrm{Co}^\gamma(x) \circ J$ for all $x \in G$ (recall that $\gamma(x, \sigma) = \gamma(x)$).

In particular, for a strongly Hamiltonian G-action the momentum map J is Co-**equivariant**, *or simply* **equivariant**, *in that $J \circ L_x = \mathrm{Co}(x) \circ J$ for all x. Moreover, infinitesimal* Co-*equivariance (in the sense that $J : S \to \mathfrak{g}^*_-$ is a Poisson map) is equivalent to global* Co-*equivariance.*

Note that (1.22) guarantees that Co^γ is an (affine) action.

Only the final claim is not immediately obvious. It is clear from (1.9) and (1.21) that the 2-cocycle Γ defined by the momentum map of the $\mathfrak{g}$-action corresponding to the group action is given in terms of γ by

$$\Gamma(X, Y) = -\frac{d}{dt}\gamma(\mathrm{Exp}(tX))(Y)_{|t=0}. \tag{1.24}$$

If we put $y = \mathrm{Exp}(tY)$ in (1.22) and differentiate with respect to t, the right-hand side vanishes when the $\mathfrak{g}$-action is strongly Hamiltonian (as $\Gamma = 0$ in that case). Hence the vanishing of the left-hand side says that $\gamma(x)$ is constant in x; since $\gamma(e) = 0$, γ identically vanishes. The equivariance is then stated by (1.21). ■

It should be remarked here that the antisymmetry of Γ as defined by (1.24) is not automatic; it is guaranteed, however, when γ is of the form (1.21).

Theorem 1.2.6. *Assume that G is simply connected, and let $\Gamma \in Z^2(\mathfrak{g}, \mathbb{R})$ and $\gamma \in Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$ be related by (1.24). There is a bijective correspondence between representations of $C^\infty_\Gamma(\mathfrak{g}^*_-)$ and Hamiltonian G-actions with 1-cocycle γ whose momentum map J is complete:*

- *Given a representation $\pi = J^* : C^\infty_\Gamma(\mathfrak{g}^*_-) \to C^\infty(S)$, define the vector fields $\xi_X := \xi_{J_X}$ (with (1.4)) on S; then the corresponding G-action exists and satisfies (1.21).*
- *Given a Hamiltonian G-action on S with momentum map J and 1-cocycle γ (defined by (1.21)), the corresponding representation π is given by $\pi = J^*$.*

This follows from Theorems 1.1.7 and 1.2.1, and the fact that γ is uniquely determined by Γ and (1.24). To see this, define $\tilde{\Gamma} : \mathfrak{g} \to \mathfrak{g}^*$ by $(\tilde{\Gamma}(X))(Y) := \Gamma(Y, X)$. This leads to an affine map $\mathrm{co}^\Gamma : \mathfrak{g}^* \to \mathfrak{g}^*$, given by $\mathrm{co}^\Gamma(X)\theta = \mathrm{co}(X)\theta + \tilde{\Gamma}(X)$. It follows from (1.11) that $\mathrm{co}^\Gamma(X)$ is a Lie algebra homomorphism; note that the Lie bracket of two affine maps $A_i = L_i + v_i$, where L is linear, is defined by $[A_1, A_2]\theta := (A_1A_2 - A_2A_1)\theta + A_1v_2 - A_2v_1$. As in the linear case, when G is simply connected there is a unique affine action $\mathrm{Co}^\gamma(G)$ on $\mathfrak{g}^*$ whose derivative is $\mathrm{co}^\Gamma(\mathfrak{g})$; it is given by (1.23), where γ satisfies (1.22), which is equivalent to (1.11). This γ then has to coincide with the same symbol defined by (1.21), stripped of its vacuous σ-dependence. ■

When $G = \tilde{G}/D$ (in the notation used after 1.2.1) is not simply connected, one has to assume integrability of the G-action. In turn, this guarantees integrability of Γ to γ; given its existence, γ is uniquely determined by the property $\tilde{\gamma} = \gamma \circ \tau_{\tilde{G}\to G}$.

Recall the definition of the space of 1-cocycles $Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$ below (1.22); define $B^1(G, \mathrm{Co}, \mathfrak{g}^*) \subset Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$ as the subspace of maps of the form

$$\gamma(x) = \mathrm{Co}(x)\theta_0 - \theta_0 \tag{1.25}$$

for some $\theta_0 \in \mathfrak{g}^*$. The 2-cocycle Γ derived from $\gamma \in B^1(G, \mathrm{Co}, \mathfrak{g}^*)$ by (1.24) lies in $B^2(\mathfrak{g}, \mathbb{R})$; it is remarkable that such a Γ is automatically antisymmetric. In general, elements of $Z^2(\mathfrak{g}, \mathbb{R})$ derived from $\gamma \in Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$ by (1.24) may fail to be antisymmetric. Elements of $Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$ that do give rise to an antisymmetric Γ are called **symplectic cocycles**, forming the space $Z^1_s(G, \mathrm{Co}, \mathfrak{g}^*)$. The **first cohomology group of G relative to the coadjoint representation** is $H^1_s(G, \mathrm{Co}, \mathfrak{g}^*) := Z^1_s(G, \mathrm{Co}, \mathfrak{g}^*)/B^1(G, \mathrm{Co}, \mathfrak{g}^*)$.

Proposition 1.2.7. *When G is simply connected, $H^2(\mathfrak{g}, \mathbb{R})$ and $H^1_s(G, \mathrm{Co}, \mathfrak{g}^*)$ are isomorphic.*

The proof of Theorem 1.2.6 shows that any $\Gamma \in Z^2(\mathfrak{g}, \mathbb{R})$ corresponds to a unique $\gamma \in Z^1(G, \mathrm{Co}, \mathfrak{g}^*)$. The claim then easily follows from the paragraph preceding the proposition. ■

More generally, further to 1.1.11 we have

Proposition 1.2.8. *When a Hamiltonian G-action with momentum map J satisfies (1.21) with (1.25), the action is strongly Hamiltonian. When $H^1_s(G, \mathrm{Co}, \mathfrak{g}^*) = 0$, any Hamiltonian action of G is strongly Hamiltonian.*

When γ is of the form (1.25), the redefined momentum map $J' = J + \theta_0$ is equivariant, as is clear from (1.21). If $H^1_s(G, \mathrm{Co}, \mathfrak{g}^*) = 0$, then any γ is of this form. ■

Note that the affine map $A : \theta \mapsto \theta + \theta_0$ satisfies $A\mathrm{Co}^\gamma(x)A^{-1} = \mathrm{Co}(x)$ for all x, as well as $\{A^*f, A^*g\}^\Gamma_\pm = A^*\{f, g\}_\pm$.

Corollary 1.2.9. *A Hamiltonian action of a compact Lie group is always strongly Hamiltonian.*

The cohomology group $H^1(G, \mathrm{Co}, \mathfrak{g}^*) := Z^1(G, \mathrm{Co}, \mathfrak{g}^*)/B^1(G, \mathrm{Co}, \mathfrak{g}^*)$ is zero when G is compact. Hence $H^1_s(G, \mathrm{Co}, \mathfrak{g}^*) \subset H^1(G, \mathrm{Co}, \mathfrak{g}^*)$ must be trivial as well. □

The group analogue of 1.1.12 is

Corollary 1.2.10. *Let $H^1(\mathfrak{g}, \mathbb{R}) = H^2(\mathfrak{g}, \mathbb{R}) = 0$, and let G act on a symplectic manifold by Poisson maps. There exists a unique equivariant momentum map J associated with this action.*

Example 1.2.11. *Let $S = T^*\mathbb{R}^n$ with its canonical symplectic structure.*

1. *The action of $G = \mathbb{R}^n$ (whose Lie algebra $\mathfrak{g} = \mathbb{R}^n$ has a basis $\{T_i\}_{i=1,\dots,n}$) on S, given by $a : (p, q) \mapsto (p, q + a)$, is Hamiltonian, with equivariant momentum map $J_i(p, q) = p_i$.*
2. *The action of $G = SO(n)$ (whose Lie algebra $\mathfrak{g} = \wedge^2(\mathbb{R}^n)$ has a natural basis $\{T_{ij}\}_{i<j=1,\dots,n}$) on S, given by $R : (p, q) \mapsto (Rp, Rq)$, is Hamiltonian, with equivariant momentum map $J_{ij}(p, q) = p_i q_j - p_j q_i$, $q_i := q^i$.*
3. *Let the abelian Lie algebra $\mathfrak{g} = \mathbb{R}^{2n}$ have a basis $\{P_i, Q^j\}_{i,j=1,\dots,n}$. The corresponding Lie group $G = \mathbb{R}^{2n}$ is parametrized by $(u, v) := \mathrm{Exp}(-uQ + vP)$; cf. II.(2.5). It acts on S by*
$$(u, v) : (p, q) \mapsto (p + u, q + v);$$
see II.(2.13), in which we have put $c = 1$. This action is Hamiltonian, with momentum map $J_{P_i}(p, q) = p_i$ and $J_{Q^i}(p, q) = q^i$. The 2-cocycle Γ is
$$\Gamma(P_i, P_j) = \Gamma(Q^i, Q^j) = 0;$$
$$\Gamma(P_i, Q^j) = -\delta_i^j. \tag{1.26}$$
The central extension $\mathfrak{g}_\Gamma$ is the Heisenberg Lie algebra $\mathfrak{h}_n$; see II.(2.4).
4. *Finally, the map J of Proposition II.3.1.4 is an equivariant momentum map for the group action II.(3.13).*

The first two examples are a special case of Lemma 2.3.1 below. As to the third, it should be remarked that since G is abelian, it must be that $B^2(\mathbb{R}^{2n}, \mathbb{R}) = 0$; cf. (1.18). Also, (1.11) is identically satisfied, so that a 2-cocycle on $\mathbb{R}^{2n}$ is simply an antisymmetric bilinear map on $\mathbb{R}^{2n}$. The dimension of the space of antisymmetric $m \times m$ matrices is $\frac{1}{2}m(m-1)$; hence $H^2(\mathbb{R}^{2n}, \mathbb{R}) = \mathbb{R}^{n(2n-1)}$.

1.3 Multipliers and Central Extensions

Definition 1.1.8 and the ensuing discussion have an analogue at the group level. The best way to approach this matter is via the following concept.